

SETTING TARGETS

● India should meet 50% of \$800 billion of chip demand through local products

● Manufacturing semiconductor products is 10 times more profitable than assembling them

NOTE OSAT IS OUTSOURCED SEMICONDUCTOR ASSEMBLY AND ATMP IS MODIFIED ASSEMBLY, TESTING, MARKING, AND PACKAGING; FIGURES IN PARENTHESES REPRESENT SHARE OF INDIAN PRODUCTS IN THE MARKET; *FIGURES FOR 2030 AND 2047 ARE ESTIMATES
SOURCE VLSI, BASED ON ANALYSIS

	2024	2030*	2047*
Total electronics market or consumption	\$ 210 bn	\$500 bn	\$3,000 bn
Domestic consumption	\$180 bn	\$320 bn	\$2,000 bn
Exports	\$30 bn	\$100 bn	\$1,000 bn
Electronics products by Indian companies	\$21 bn (10%)	\$125 bn (25%)	\$1,500 bn (50%)
Electronics products companies	200	1,000	5,000
Total semiconductor market/consumption	\$40 bn	\$135 bn	\$800 bn
No. of wafer fabs in operation	N/A	5	20
No. of OSAT/ATMP plants in operation	N/A	10	20
Semiconductor product companies	50	200	500
Number of Unicorns in electronics, semiconductors	N/A	10	50
Employment-generation targets	N/A	12.5 mn	100 mn

related services to the semiconductor industry, emphasises that this is a global issue. There is a worldwide talent shortage that might hinder the ability to meet the anticipated AI-driven demand for chips, and educating the next-generation of engineers is challenging, as providing physical access to the most advanced nanotechnologies is expensive for academic institutions. He believes the solution lies in the virtual simulation of real-world labs.

To address the talent gap, the government is rolling out initiatives aimed at skilled workers. Lam Research has signed an MoU with the India Semiconductor Mission and IISc to train more than 60,000 engineers in semiconductor manufacturing technology.

VLSI Society has started an undergraduate programme in VLSI design in 350 BTech colleges, 450 colleges offering Masters and PhD programmes. Still, it is a struggle to find trained faculty and infrastruc-

ture in terms of design tools.

Rabindra Srikantan, the Vice Chairman of CII Karnataka and Founder and Managing Director of technological services firm ASM Technologies Ltd, says some states like Karnataka have taken the lead in developing skill clusters, especially in semiconductors.

TOWARDS 2047

India has a robust chip design industry and it must establish similar dominance in chip manufacturing. To further develop the domestic value chain, the centre is expected to introduce a components incentive scheme. Make in India has a unique opportunity to build for and take advantage of the Indian market, says Singh. "So, I think there's a huge opportunity ahead of us and Micron is well-positioned to support our customer base," he adds.

Experts say the question now is how India can convert these grow-

ing semiconductor consumption numbers to local demand and for local products.

If India is targeting \$100 billion in demand by 2030, it must reach the goal of ensuring at least 30% of this consumption is fulfilled by Indian products. "It will increase the contribution to GDP. We need to make more economic impact by moving to products," says VLSI Society's Gupta.

In Gupta's opinion, the business side of the semiconductor industry will be the biggest challenge in the years to come. He feels the government could drive domestic demand through mandates for 'Make In India' chips.

India has caught global attention as it treads the path to becoming a semiconductor hub. But will that be enough to scale domestic companies up to the levels of AMD or Nvidia? Only time will tell. **BT**

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